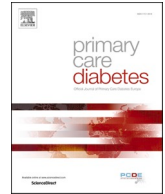




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Comparative assessment of diagnostic agreement between artificial intelligence and general practitioners in diabetic retinopathy screening using non-mydriatic fundus photography

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ABSTRACT

Aims: To assess the diagnostic agreement between an artificial intelligence (AI) system and general practitioners (GPs) interpreting fundus photographs for diabetic retinopathy (DR) screening, using the ophthalmologists' assessment as the reference standard.

Methods: We performed a cross-sectional study of 500 primary care patients with type 2 diabetes (T2DM). Each underwent two 45° non-mydriatic fundus photos per eye. Images were independently evaluated by three trained GPs, a deep learning-based AI system (EyeArt v3.0.0 (Eyenuk Inc.)), and an ophthalmologist (reference standard). Diagnostic agreement was measured with Cohen's kappa and Cramer's V, and sensitivity, specificity, predictive values, likelihood ratios, and overall accuracy were calculated.

Results: Mean age was 64 years, and 59% were men. DR prevalence was 11%. AI showed near-perfect agreement with ophthalmology ($\kappa = 0.91$) and outperformed GPs, with 98.2% sensitivity, 98.9% specificity, and 98.8% accuracy. GPs showed substantial agreement ($\kappa = 0.84$), with lower sensitivity (86.4%) but similarly high specificity (98.1%). In multinomial analysis, AI achieved 87.6% sensitivity for mild DR and 100% for moderate-to-severe DR, missing no clinically relevant cases.

Conclusions: In this cohort restricted to gradable images, AI demonstrated higher sensitivity, specificity, and diagnostic agreement than GPs when compared with a single ophthalmologist reference standard. Supervised use in primary care could strengthen population-based screening, but large-scale adoption will require multicentre studies, formal cost-effectiveness analyses, and validation in unselected screening populations.

1. Aims

Diabetes mellitus (DM) is a major global epidemic, affecting 589 million adults worldwide, with projections rising to 853 million by 2050 [1]. In Spain, its age-adjusted prevalence is around 10%, one of the highest in Europe [1]. Among its microvascular complications, diabetic retinopathy (DR) is both the most common and the leading cause of preventable visual loss in working-age adults [2], affecting 23% of people with DM—over 100 million cases globally [3]. DR progresses from mild non-proliferative stages to advanced proliferative disease, making early detection key to preventing irreversible visual damage [4].

Systematic screening reduces visual loss by enabling early detection and timely treatment of retinal lesions [5]. However, the most accurate

techniques—such as seven-field fundus photography or slit-lamp bio microscopy—are difficult to implement at a population level, as they require specialised equipment and highly trained ophthalmologists [5].

In clinical practice, non-mydriatic digital retinal photography has become the preferred screening tool in primary care due to its accessibility and effectiveness [5]. It is often used alongside tele-ophthalmology, which broadens coverage and enhances the overall efficiency of screening programmes [5].

Multicentre reviews and our RedETS/AQuAS national report support its inclusion in population-based screening programmes, citing clear benefits in accessibility, cost-effectiveness, and visual outcomes [6,7]. However, image interpretation in primary care requires specific training and can vary between professionals, which may lead to diagnostic

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inconsistencies and referral delays.

The integration of artificial intelligence (AI) systems, particularly deep-learning algorithms trained on large datasets of retinal images, has marked a significant step forward in DR screening [8]. Recent studies show that these systems can perform at least as well as, and in some cases better than, ophthalmologists and trained general practitioners (GPs) [9]. Their potential advantages include immediate results, reduced clinical workload, and the feasibility of conducting large-scale screening without the need for initial image grading by clinicians [5]. However, evidence on real-world effectiveness, agreement with GPs, cost-effectiveness and organisational impact remains limited [10].

Since population-based screening largely occurs in primary care, understanding how well AI systems perform in this setting, where GPs have a central role, is essential. However, we found no studies that specifically assess the diagnostic agreement between AI and GPs using ophthalmological assessment as the reference standard. This lack of evidence highlights the need for real-world studies in which AI is used as a support tool within screening pathways.

This study assessed the hypothesis that the diagnostic agreement between AI and trained GPs is at least comparable to that achieved against the ophthalmological reference standard and evaluated diagnostic performance and organisational implications for primary care-based screening.

2. Methods

2.1. Study design

We conducted a multicentre, retrospective observational study using consecutive cohorts from the primary care screening programme for DR.

The study was carried out in the Zaragoza II healthcare sector between June and July 2025, involving 21 primary care centres and a population of ~250,000 residents.

The study objectives were:

1. Assess the diagnostic agreement between an AI system and trained GPs interpreting retinal photographs, using ophthalmology as the reference standard (Cohen's Kappa).
2. Examine diagnostic performance (sensitivity, specificity, predictive values, overall accuracy and likelihood ratios).
3. Explore the potential impact of AI on screening efficiency and follow-up pathways for DR in primary care and assess the feasibility of a hybrid AI-GP model as a screening strategy in this setting.

2.2. Population and sample

We recruited 500 consecutive patients with type 2 diabetes (T2D) attending routine primary care appointments, following their standard appointment order to minimise selection bias. Eligible participants were adults (≥ 18 years) with a confirmed diagnosis of T2D; only cases of gestational diabetes were excluded.

The sample size was calculated using the standard formula for a single proportion, assuming a 15% prevalence of DR, a 95% confidence level and a 3% absolute precision, yielding a minimum requirement of 490 patients. Including 500 participants ensured adequate statistical power.

The primary unit of analysis was the eye ($n = 1000$), with an additional patient-level analysis based on the presence of DR in either eye.

2.3. Diagnostic procedure

The screening protocol followed current recommendations within the Spanish national health system: a two-year review interval for individuals without DR, annual screening for those with mild DR, and referral to ophthalmology for moderate or severe cases [5].

Each patient underwent two fundus photographs per eye—one

centred on the macula and one on the optic nerve—using a 45° non-mydratric camera with $\geq 80\%$ sensitivity, $\geq 95\%$ specificity and a technical failure rate below 5% (TOPCON TRC NW-400). Image resolution was sufficient to assess the central 40° of the retinas. The final diagnosis for each eye was based on the combined assessment of both images. All fundus photographs were independently evaluated by three sources:

1. A group of three GPs, each with approximately 15 years of experience in interpreting fundus images. GP readers (comparator) — Training and experience were documented using program records when available. Where records were incomplete, we collected standardized self-reported information using a prespecified structured questionnaire (Supplementary Table S2), including: years interpreting retinal images, training components and duration (hours), number of supervised cases, assessment/feedback procedures, and ongoing quality assurance/audit frequency. Annual reading volume was extracted from program logs when available; otherwise it was estimated using a prespecified approach (screening sessions/week $\times$ mean examinations/session $\times$ years in role) and cross-checked against local screening activity reports.

Note: Training hours, supervised cases, and monthly reading time are standardized self-reported estimates from a structured questionnaire when complete programme records were unavailable. “Supervised cases” refers to onboarding supervision, not ongoing monthly double-reading.

2. A deep learning AI system designed for the automated screening of DR.
3. A retinal specialist ophthalmologist (reference standard). The reference classification was provided by a single retinal specialist ophthalmologist; no double reading or adjudication process was performed. Intra-observer variability was not formally assessed in this study and is acknowledged as a limitation.

Cases were evenly distributed among the three GPs, so that each image was assessed by a single GP rather than being reviewed by all of them.

Ophthalmology's diagnosis served as the reference for validating GP readings and guiding management.

2.4. Artificial intelligence

Retinal images were analyzed using the Eyenuk EyeArt system (software version v3.0.0 Rev.101, 2024), deployed in its standard clinical configuration. The system evaluates fundus images and generates an automated eye-level classification when image quality is deemed sufficient by its internal gradability module.

Outputs were prespecified and mapped to analytic categories: no diabetic retinopathy (DR), mild non-proliferative DR, and moderate-to-severe or proliferative DR, consistent with the International Clinical Diabetic Retinopathy scale. Proprietary details regarding model architecture, training datasets, and operating thresholds are not publicly disclosed; therefore, the system was used with its default locked clinical operating configuration. External validation of the CE-marked system has been reported in independent real-world studies [11,12].

According to manufacturer documentation, EyeArt combines outputs from multiple deep neural networks within a clinically aligned framework. The manufacturer reports training on $> 375,000$ expert-graded images and validation on $> 250,000$ images from geographically diverse screening programmes, covering a wide range of image quality and demographics. Under EU MDR, the system is registered as a Class IIb medical device, and the instructions for use recommend referral for positive and ungradable results.

When image quality is insufficient for classification, the system produces an explicit “ungradable” output. Such cases are managed according to programme protocol, including repeat imaging and/or

clinician review with referral as appropriate.

The analytic cohort was restricted to examinations with complete AI outputs for both eyes; therefore, all included cases were gradable by design. Real-world rates of non-gradable examinations cannot be inferred from this dataset.

The software was CE-marked and implemented within the screening pathway as an automated DR classification tool supporting repeat imaging and referral decisions according to programme protocol.

2.5. Variables

We collected sociodemographic and clinical variables, including age, sex, age at diagnosis, and duration of T2D. DR severity was classified according to the International Clinical Diabetic Retinopathy Disease Severity Scale into three categories: no DR, mild non-proliferative DR, and moderate-to-severe or proliferative DR.

For the analysis, we conducted both dichotomous comparisons (DR present/absent) and multinomial assessments based on the three severity categories. All readers were fully blinded: the ophthalmologist was unaware of the readings made by the AI system and the GPs; the GPs did not have access to the AI outputs or the ophthalmologist's grading; and the AI system operated entirely independently. The study design, data collection, and reporting adhered to the STARD guidelines for diagnostic accuracy research.

2.6. Ethical considerations

The study was conducted per the Declaration of Helsinki, and data were anonymised before analysis in compliance with the EU General Data Protection Regulation (GDPR).

The protocol was approved by the Aragón Research Ethics Committee (CEICA, reference PI24/502). No individual consent was required due to the retrospective design, but written consent was obtained from GPs and the ophthalmologist for the use of their assessments in the comparative analysis.

2.7. Statistical analysis

Because two eyes per patient were included, eye-level analyses are subject to within-patient correlation. Confidence intervals were calculated under eye-level independence assumptions and may therefore be anti-conservative (i.e., too narrow). Future confirmatory studies should use clustered inference (e.g., generalized estimating equations, cluster-robust variance estimators, or patient-level bootstrap resampling) to fully account for within-patient clustering.

All statistical analyses were conducted using R software (version 4.3.0) [13], using standard packages for diagnostic test accuracy and agreement statistics.

3. Results

3.1. Descriptive of the Study Sample

A total of 500 patients with T2D were included, contributing 1,000 eyes (two per patient). Of these, 59% were men, with a mean age of 64 years (SD 11; range 58–72). The mean age at diagnosis was 55 years (SD 11), and the average duration of the disease was 9 years (SD 6).

3.2. Diagnostic distribution

According to the ophthalmological reference diagnosis, 89% of eyes showed no DR, 8.9% had mild non-proliferative DR, and 2.1% had moderate-to-severe disease.

The AI system classified 88% of eyes as no DR, 8.8% as mild DR, and 3.2% as moderate-to-severe DR. GPs classified 89% as no DR, 8.4% as mild DR, and 2.6% as moderate-to-severe DR.

Gradability / image quality — In the analytic cohort ($n = 500$ patients; $n = 1000$ eyes), the AI system returned a diagnostic output for all eyes (100% gradable by design), as the dataset was restricted to examinations with complete AI output. Therefore, the real-world proportion of non-gradable images cannot be inferred from this study sample.

3.3. Diagnostic agreement

3.3.1. AI vs Ophthalmology

Agreement between AI and ophthalmology was almost perfect (Cohen's Kappa = 0.907; Cramer's V = 0.850). The AI correctly identified 98.9% of cases without DR, 87.6% with mild DR, and 100% with moderate-to-severe DR. Most discrepancies occurred in mild DR, which the AI sometimes labelled as no DR, but no moderate-to-severe case was missed. [Table 1](#)

3.3.2. GPs vs. Ophthalmology

Agreement between GPs and ophthalmology was high (Kappa = 0.837; Cramer's V = 0.760). GPs correctly identified 98.1% of cases without DR, 79.8% of mild DR cases and 76.2% of moderate-to-severe DR cases. Discrepancies mainly involved underdiagnosis of mild and moderate cases, frequently classified as no DR, resulting in lower sensitivity. [Table 2](#)

3.3.3. GPs vs. AI

Agreement between GPs and AI was strong (Kappa = 0.775; Cramer's V = 0.705). GPs correctly identified 97.4% of cases without DR, 74.2% of mild DR cases, and 78.3% of moderate-to-severe DR cases. Most discrepancies were in mild DR, with both under- and over-estimation. [Table 3](#)

3.4. Diagnostic performance

The AI system delivered consistently stronger diagnostic performance than the GPs when compared with the ophthalmological reference standard, demonstrating higher sensitivity, specificity, and overall accuracy ([Table 4](#)).

3.4.1. Dichotomous analysis (DR present vs. absent)

DR prevalence was 11%. AI classified 11.8% of eyes as DR, and GPs, 11.2% of them.

AI achieved a sensitivity of 98.2%, specificity of 98.9%, an overall accuracy of 98.8%, a negative predictive value (NPV) of 99.7%, and a negative likelihood ratio (LR⁻) of 0.02. GPs reached a sensitivity of 86.4%, specificity of 98.1%, overall accuracy of 96.8%, an NPV of 98.7%, and an LR⁻ of 0.14.

Differences in sensitivity and overall accuracy between AI and GPs were statistically significant ($p < 0.01$), reinforcing the superior

Table 1

Diagnostic agreement between the AI system and ophthalmology (reference standard) in DR classification. Notes: $\chi^2 = 1445.8$; $df = 4$; $p < 0.001$; Cramer's V = 0.850; Cohen's Kappa = 0.907; 95% CI: 0.851–0.949; Fisher's exact test. Abbreviations: AI = artificial intelligence; DR = diabetic retinopathy; No DR = no diabetic retinopathy.

Ophthalmology diagnosis	AI: No DR (n, %)	AI: Mild DR (n, %)	AI: Moderate-to-severe DR (n, %)	Total (n, %)
No DR	880 (98.9%)	6 (0.7%)	4 (0.4%)	890 (100%)
Mild DR	2 (2.2%)	78 (87.6%)	9 (10.1%)	89 (100%)
Moderate-to-severe DR	0 (0.0%)	0 (0.0%)	21 (100%)	21 (100%)
Total	882 (88.2%)	84 (8.4%)	34 (3.4%)	1000 (100%)

Table 2

Diagnostic agreement between GPs and ophthalmology (reference standard) in DR classification. Notes: $\chi^2 = 1154.3$; $df = 4$; $p < 0.001$; Cramer's $V = 0.760$; Kappa = 0.837; 95% CI: 0.774–0.892; Fisher's exact test. Abbreviations: DR = diabetic retinopathy; No DR = no diabetic retinopathy; GPs = general practitioners.

Ophthalmology diagnosis	GPs: No DR (n, %)	GPs: Mild DR (n, %)	GPs: Moderate-to-severe DR (n, %)	Total (n, %)
No DR	873 (98.1%)	13 (1.5%)	4 (0.4%)	890 (100%)
Mild DR	15 (16.9%)	71 (79.8%)	3 (3.4%)	89 (100%)
Moderate-to-severe DR	0 (0.0%)	5 (23.8%)	16 (76.2%)	21 (100%)
Total	888 (88.8%)	89 (8.9%)	23 (2.3%)	1000 (100%)

Table 3

Diagnostic agreement between GPs and the AI system in DR classification. Notes: $\chi^2 = 993.9$; $p < 0.001$; Cramer's $V = 0.705$; Kappa = 0.775; 95% CI: 0.702–0.843. Abbreviations: DR = diabetic retinopathy; No DR = no diabetic retinopathy; GPs = general practitioners.

GPs diagnosis	AI: No DR (n, %)	AI: Mild DR (n, %)	AI: Moderate-to-severe DR (n, %)	Total (n, %)
No DR	865 (97.4%)	17 (1.9%)	6 (0.7%)	888 (100%)
Mild DR	13 (14.6%)	66 (74.2%)	10 (11.2%)	89 (100%)
Moderate-to-severe DR	4 (17.4%)	1 (4.3%)	18 (78.3%)	23 (100%)
Total	882 (88.2%)	84 (8.4%)	34 (3.4%)	1000 (100%)

Table 4

Diagnostic performance and agreement between GPs and the AI system in DR classification compared with ophthalmology (reference standard). Diagnostic accuracy metrics (sensitivity, specificity, predictive values, Youden's index and likelihood ratios) were calculated only against the ophthalmological assessment, which served as the reference standard. For the comparison between GPs and AI, no independent reference standard exists; therefore, only agreement measures (e.g., Kappa, Cramer's V) are reported. Abbreviations: PPV = positive predictive value; NPV = negative predictive value; LR+ = positive likelihood ratio; LR– = negative likelihood ratio; 95% CI = 95% confidence interval.

Metrics	AI vs. Ophthalmology	GPs vs. Ophthalmology	GPs vs. AI
Sensitivity (%)	98.2 (95% CI: 93.6–99.8)	86.4 (95% CI: 78.5–92.2)	–
Specificity (%)	98.9 (95% CI: 97.9–99.5)	98.1 (95% CI: 97.0–98.9)	–
PPV (%)	91.5 (95% CI: 85.0–95.9)	84.8 (95% CI: 76.8–90.9)	–
NPV (%)	99.8 (95% CI: 99.2–100)	98.3 (95% CI: 97.2–99.1)	–
Overall accuracy (%)	98.8 (95% CI: 97.9–99.4)	96.8 (95% CI: 95.5–97.8)	–
Youden's index	0.97	0.85	–
Kappa (95% CI)	0.907 (0.851–0.949)	0.837 (0.774–0.892)	0.775 (0.702–0.843)
Cramer's V	0.850	0.760	0.705
LR+	87.4	45.2	–
LR–	0.02	0.14	–

diagnostic performance of the AI system. (Table 4)

Supplementary Table S4. Discordant classifications between AI and GP readers (eye-level; $n = 1000$ eyes).

Supplementary Table S5. Patient-level diagnostic performance

($n = 500$ patients; positive if either eye has DR) against the ophthalmologist reference standard.

3.4.2. Multinomial analysis (No DR / Mild DR / Moderate–severe DR)

The AI system achieved a sensitivity of 87.6% for mild DR and 100% for moderate-to-severe DR, with overall accuracy exceeding 90% across all categories. GPs showed lower sensitivities for both mild (79.8%) and moderate-to-severe cases (76.2%), with a greater tendency to misclassify mild DR as “No DR”.

3.5. Overall comparison

Overall, the highest level of agreement was observed between the AI system and the ophthalmological diagnosis, with an almost perfect agreement, as shown in Fig. 1.

Comparison	No DR	Mild DR	Moderate-to-severe DR
AI vs Ophthalmology	98.9	87.6	100.0
GPs vs Ophthalmology	98.1	79.8	76.2
GPs vs AI	97.4	74.2	78.3

3.6. Sub-analysis

No significant differences in diagnostic agreement were observed according to sex, age, or duration of diabetes.

3.7. Invalid images

The analytic cohort included only examinations with a gradable AI output (100% gradable by design), as the dataset was restricted to eyes with complete DR severity classification available for both eyes. GPs issued a diagnosis for all images included in the analytic cohort. In routine practice, the system may return an explicit “ungradable” output; these encounters follow the standard pathway (repeat imaging and/or clinician review with referral as per programme protocol), but were excluded prior to analytic cohort assembly.

4. Discussion

AI demonstrated diagnostic accuracy comparable to that of the ophthalmologist and showed strong agreement with GPs, supporting its potential use as a reliable tool for population-based screening under clinical supervision. Most discrepancies arose in mild and moderate cases, a pattern well described in the literature, highlighting the challenges of grading early-stage DR for both algorithms and human readers [14].

The diagnostic performance indicators confirmed the clear superiority of AI over GPs in sensitivity, specificity and predictive values, with an overall accuracy near 99%. These findings are consistent with international evidence, including the pivotal IDX-DR trial—the first FDA-approved autonomous screening system—which reported performance comparable to that of retinal specialists [14]. The high sensitivity and strong NPV observed in our study suggest that AI may be particularly valuable for ruling out disease in at-risk populations, shortening reporting times and reducing ophthalmology workload.

Meanwhile, GPs showed strong diagnostic performance, with high specificity and overall accuracy, reinforcing their role in primary care-based screening and their potential synergy with AI systems. Our findings add to evidence supporting hybrid models—in which AI acts as an automated filter and the GPs review and manage positive cases—potentially improving screening coverage and adherence [15]. Progressive integration of AI-assisted reading, supported by feedback and validation, could further enhance diagnostic quality and patient safety.

In Spain, evidence on the use of AI in population-based DR screening programmes is still limited, although reports from real-world

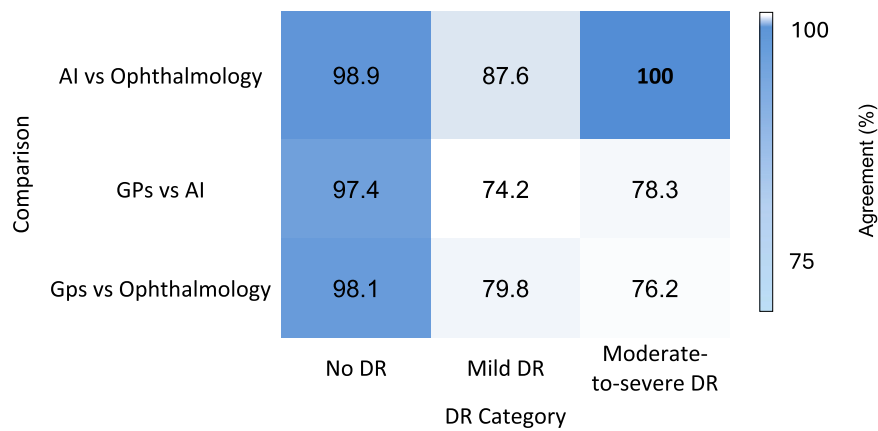


Fig. 1. Diagnostic agreement by category (%) between AI, GPs and ophthalmology. Table showing agreement percentages by DR category for the three comparisons; numeric values are displayed to improve accessibility in monochrome/print viewing. Abbreviations: AI = artificial intelligence; GPs = general practitioners; DR = diabetic retinopathy.

experiences are emerging. A recent study validated an automated system in routine practice, showing high diagnostic performance and results consistent with international research [16]. These findings support the external validity of our study and its potential applicability to national and international programmes. Beyond technical performance—and although our study did not assess organisational or acceptability aspects—the literature highlights the importance of ethical and operational considerations, including patient and professional acceptance, regulation of automated processes, and the challenges of large-scale implementation [17].

In line with international findings, AI showed performance comparable to other healthcare professionals trained in fundus image interpretation [18], supporting its potential as a complementary tool within screening programmes. National data also indicate that an AI-based model in primary care is feasible and can expand screening in a cost-effective way [19].

Our results support integrating AI with primary care to streamline DR screening through a simple routine: fundus photography, immediate AI interpretation and management based on the result. Under this model, negative cases continue routine primary care follow-up, while only positive or non-gradable images are referred to ophthalmology. This approach reduces unnecessary referrals, enables same-day results and increases coverage while keeping GPs central to decision-making.

4.1. Strengths

This study has a strong design since it used the ophthalmologist's diagnosis as the reference standard, with an adequate sample size and rigorous analyses ensuring internal validity. A key strength is the comparator: GPs specifically trained in fundus interpretation, offering a realistic and demanding benchmark. It is also the first Spanish study assessing combined AI–GP performance in a population-based DR screening programme within the public health system under real-world conditions. The findings highlight how technology can complement—rather than replace—clinical judgement, particularly relevant in European primary care-centred health systems.

4.2. Limitations

Discrepancies in mild and moderate cases underscore the difficulty for both AI and GPs in detecting early stages of and highlight the DR need for more robust models and improved training. Conducting the study in a single healthcare sector limits generalisability. The 11% prevalence may have affected predictive values, and restricting the analytic cohort to examinations with complete AI output (100% gradable by design) may have overestimated performance compared with routine

practice where a proportion of examinations may be labeled non-gradable and require repeat imaging and/or clinical review. Other known limitations include limited validation across diverse populations, restricted explainability and the inability of some systems to detect other ocular conditions such as glaucoma or AMD [18,19]. Multicentre studies and the inclusion of an expert adjudication committee would help strengthen external validity.

Additional limitations relevant to reproducibility include: (i) the reference standard was provided by a single retinal specialist, without double reading or adjudication, and intra-observer variability was not assessed; (ii) manufacturer-disclosed details regarding AI training data, labeling pipeline, and proprietary operating thresholds were not accessible; therefore, we report the exact software version, deployed operating configuration, and available external validation evidence; (iii) GP training and experience were documented from program records when available, and otherwise relied on standardized self-report, which may introduce reporting imprecision; and (iv) eye-level analyses may be influenced by within-patient correlation. Importantly, all diagnostic accuracy estimates represent agreement with a single expert clinical reader rather than an infallible ground truth; future studies should incorporate independent double grading and/or a consensus adjudication panel (≥ 2 –3 expert graders) and consider intra-observer repeat grading.

4.2.1. Clinical and Public Health Implications

The observation that AI did not miss any moderate-to-severe cases has clear clinical significance, as it ensures that the highest-risk patients would not go undetected in automated screening. This reinforces its value as an initial filter, prioritising referrals and enabling ophthalmologists to focus on complex cases without compromising diagnostic quality.

Within primary care, an AI–GPs hybrid model offers an opportunity to decentralise screening, expand coverage and improve coordination across care levels, particularly in the context of the steadily increasing burden of diabetes and DR.

Immediate AI results may enhance better follow-up adherence and reduce geographical inequities by bringing early detection closer to rural or specialist-scarce areas [20]. From a population perspective, such an approach could increase screening uptake and reduce preventable diabetes-related blindness.

From an organisational and economic perspective, integrating AI into screening programmes represents a step forward that may improve efficiency and optimise specialist resources. Evidence suggests that, although initial investment is required, AI-supported screening may be cost-effective in the long term by preventing DR complications [21]. However, widespread implementation requires sustainable funding,

regulatory frameworks and consistent clinical oversight.

In low-prevalence screening settings, predictive values are highly prevalence-dependent. Even with a high PPV, a small false-positive proportion can translate into a meaningful number of unnecessary referrals at scale; therefore, local prevalence and service capacity should be explicitly incorporated into implementation planning and cost-effectiveness models.

4.2.2. Future perspectives

The future of AI in DR screening depends on ongoing validation, interoperability with electronic health records and the development of explainable algorithms that strengthen trust among both professionals and the public [22]. Training healthcare teams and ensuring transparency will be essential for safe and effective adoption. Multicentre, prospective studies are needed to confirm external validity and explore innovations such as ophthalmology, which positions the retina as a systemic biomarker [28]. In addition, the black-box nature of many deep-learning systems has medico-legal and trust implications: when errors occur, the lack of an interpretable rationale can hinder clinician oversight. Implementation should therefore include clear human-oversight responsibilities, training to mitigate automation bias, and quality assurance processes to support appropriate clinician challenge of AI outputs.

Priority areas include validating AI in different clinical settings, conducting implementation trials (AI + GPs vs GPs + tele-ophthalmology), exploring perceptions and barriers among professionals and patients, and extending detection to other ocular conditions such as age-related macular degeneration or glaucoma. It will also be important to assess hybrid AI-GP models, their longitudinal impact and their cost-effectiveness in real-world settings.

Important questions remain unanswered: the optimal degree of human oversight, how to assess confidence in AI-assisted decisions, and how to ensure the continuous updating of algorithms in routine practice. A rigorously validated and ethically integrated AI system has the potential to become a transformative tool for DR screening and for preventing avoidable blindness, supporting more equitable and efficient care across primary care settings.

5. Conclusions

AI-based DR screening showed almost perfect agreement with ophthalmologists.

GPs achieved substantial agreement, although with lower sensitivity than AI.

AI did not miss any moderate-to-severe DR cases in this real-world screening programme.

A supervised hybrid AI-primary care model could improve the efficiency of DR screening.

Our findings support this integration as a safe and effective strategy, although widespread adoption will require multicentre prospective studies and formal cost-effectiveness analyses to confirm its feasibility and long-term sustainability.

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Declaration of Generative AI and AI-assisted technologies in the writing process

The authors used ChatGPT (OpenAI) solely for language support, content organisation and text refinement. All generated content was subsequently reviewed, edited and thoroughly verified by the authors, who take full responsibility for the final version of the manuscript.

Declaration of Competing Interest

The lead author declares no conflicts of interest. All remaining co-authors likewise report no conflicts of interest related to this study.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.pcd.2026.04.007](https://doi.org/10.1016/j.pcd.2026.04.007).

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